

# The Crime Lab | CyberDefenders Writeup

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[Practice](#) > [SOC Analyst Tier 1](#) > [Level 2](#) > [The Crime](#)

## The Crime Lab

Utilize ALEAPP to analyze Android device artifacts, reconstructing a victim's financial details, movements, and communication patterns.

Category: [Endpoint Forensics](#)

Tools: [ALEAPP](#) [DB Browser for SQLite](#)

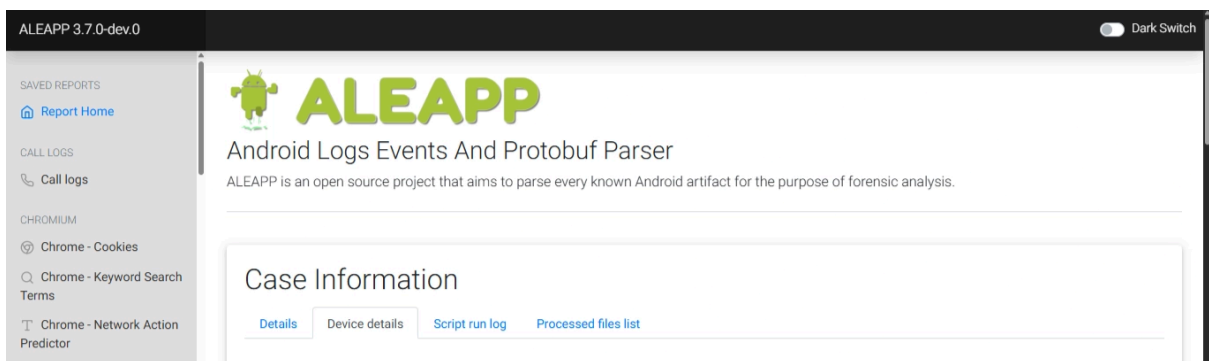
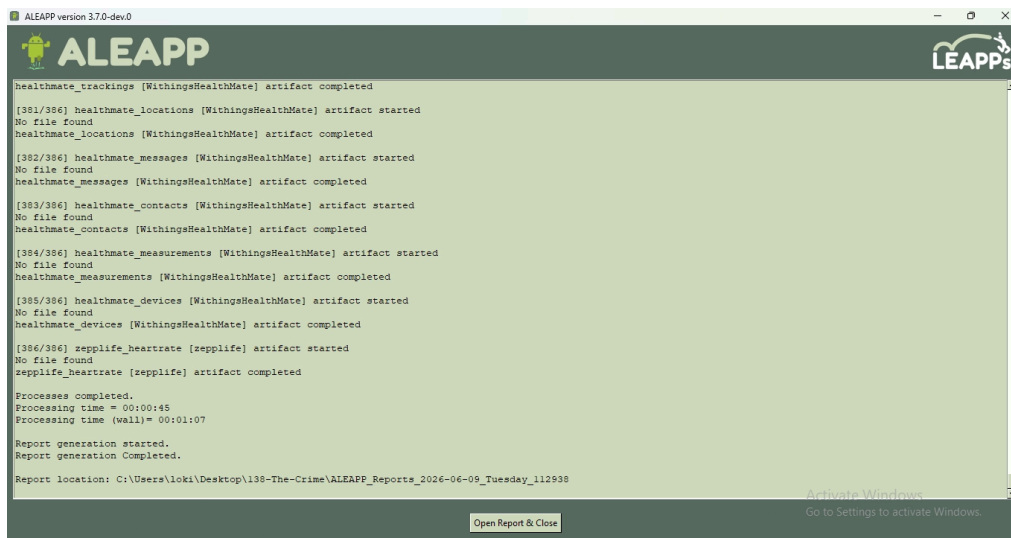
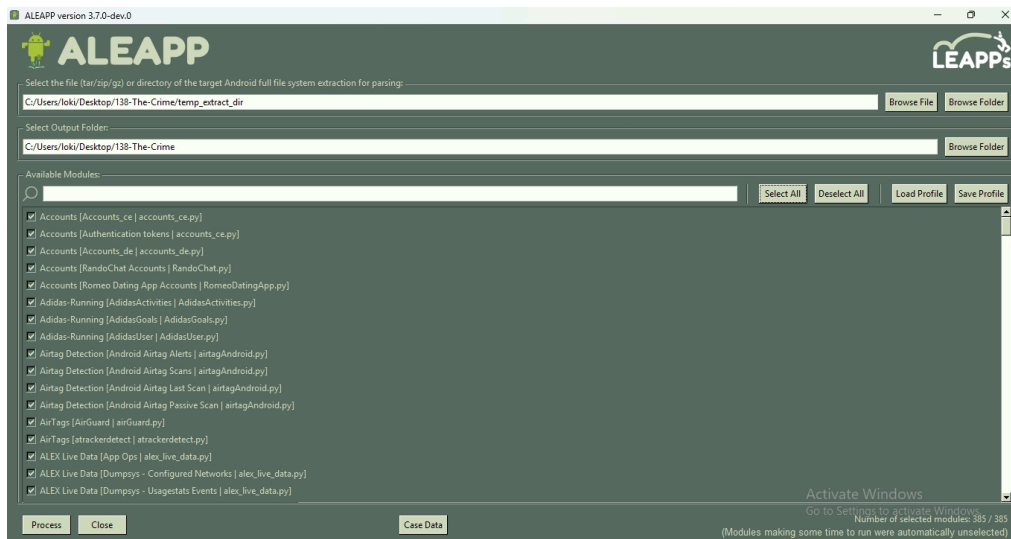
🔧 Easy   ● Retired   ⌚ 1hr   ★★★★★ 4.6

☰ Scenariod

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We're currently in the midst of a murder investigation, and we've obtained the victim's phone as a key piece of evidence. After conducting interviews with witnesses and those in the victim's inner circle, your objective is to meticulously analyze the information we've gathered and diligently trace the evidence to piece together the sequence of events leading up to the incident.

setup:

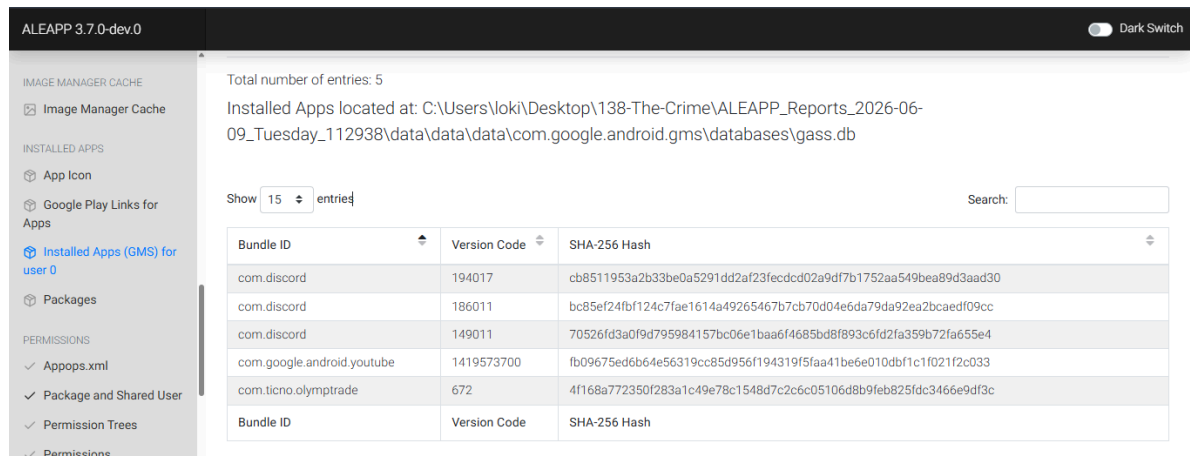


Q1

Based on the accounts of the witnesses and individuals close to the victim, it has become clear that the victim was interested in trading. This has led him to invest all of his money and acquire debt. Can you identify the **SHA256** of the trading application the victim primarily used on his phone?

**process :**

As requested, we identified the trading application used by the victim within the installed applications section. Alongside the application entry, its SHA-256 hash is also displayed.



**flag:4f168a772350f283a1c49e78c1548d7c2c6c05106d8b9feb825fdc3466e9df3c**

Q2

According to the testimony of the victim's best friend, he said, "While we were together, my friend got several calls he avoided. He said he owed the caller a lot of money but couldn't repay now". How much does the victim owe this person?

**process :**

We then proceeded to the SMS messages section and, based on the conversations present, identified the relevant information required for the investigation.

| Date                   | MSG ID | Thread ID | Address       | Contact ID | Date sent              |
|------------------------|--------|-----------|---------------|------------|------------------------|
| 2023-09-20<br>20:09:49 | 1      | 3         | +201172137258 | 6          | 2023-09-20<br>21:09:49 |

The victim received a message from the caller threatening them to pay 250,000 EGP, which the caller claimed was owed by the victim.

ALEAPP 3.7.0-dev.0

GOOGLE PHOTOS

Google Photos (gphotos1) - Cache

IMAGE MANAGER CACHE

Image Manager Cache

INSTALLED APPS

App Icon

Google Play Links for Apps

Installed Apps (GMS) for user 0

Packages

PERMISSIONS

Appops.xml

Package and Shared User

Permission Trees

Permissions

Runtime Permissions 0

POWER EVENTS

Last Boot Time

RECENT ACTIVITY

Recent Activity 0

BUGS & MMS

SMS messages

SQLite JOURNALING

<

flag:250000

Q3

What is the name of the person to whom the victim owes money?

process :

Referring back to the earlier conversation, we observed that the “address” field appears to contain a phone number, as shown in the image.

| Data 1           | Display Name | Phone Number     |
|------------------|--------------|------------------|
| +20 117 213 7258 | Shady Wahab  | +20 117 213 7258 |

ALEAPP 3.7.0-dev.0

Chrome - Network Action Predictor

Chrome - Search Terms

Chrome - Web History

Chrome - Web Visits

CONTACTS

Contacts

DEVICE INFO

Partner Settings

SIM info 0

Settings Secure 0

DEVICE INFORMATION

ADB Hosts

OS Version

ALEAPP

Contacts report

Contacts from the device

Total number of entries: 6

Contacts located at: See source file(s) below

Show

15

entries

Search:

| Mimetype                         | Data 1           | Display Name | Phone Number     | Email Address | Source File                                                                                                                     |
|----------------------------------|------------------|--------------|------------------|---------------|---------------------------------------------------------------------------------------------------------------------------------|
| vnd.android.cursor.item/phone_v2 | +20 117 213 7258 | Shady Wahab  | +20 117 213 7258 |               | \\?C:\Users\loki\Desktop\138-The-Crime\ALEAPP_Reports_2026-06-09_Tuesday_112938\data\data\data\com.android.providers.contacts\c |
| vnd.android.cursor.item/phone_v2 | +20 117 758 0642 | Mark         | +20 117 758 0642 |               | \\?C:\Users\loki\Desktop\138-The-Crime\ALEAPP_Reports_2026-06-09_Tuesday_112938\data\data\data\com.android.providers.contacts\c |

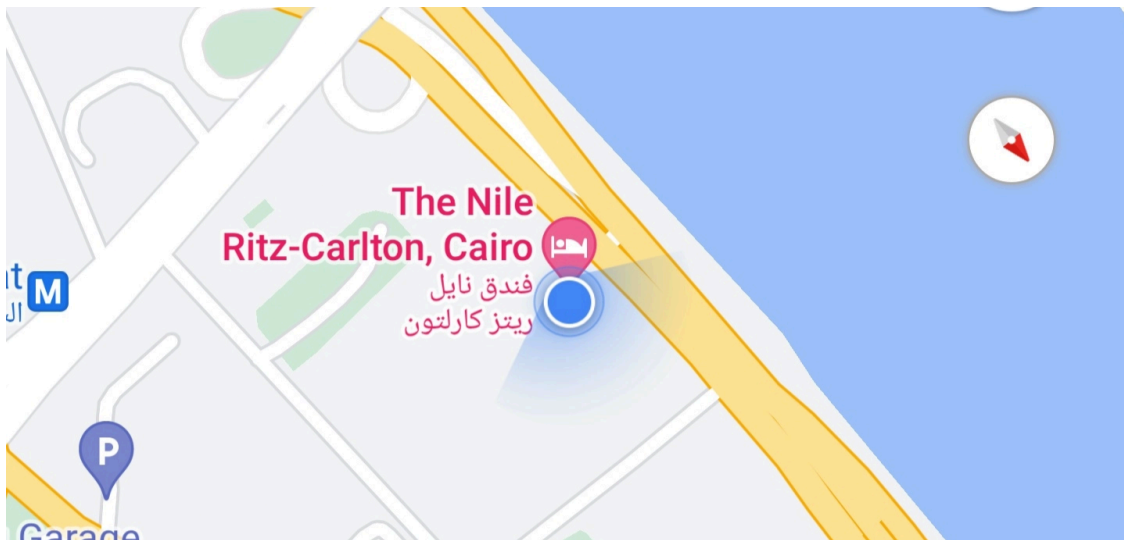
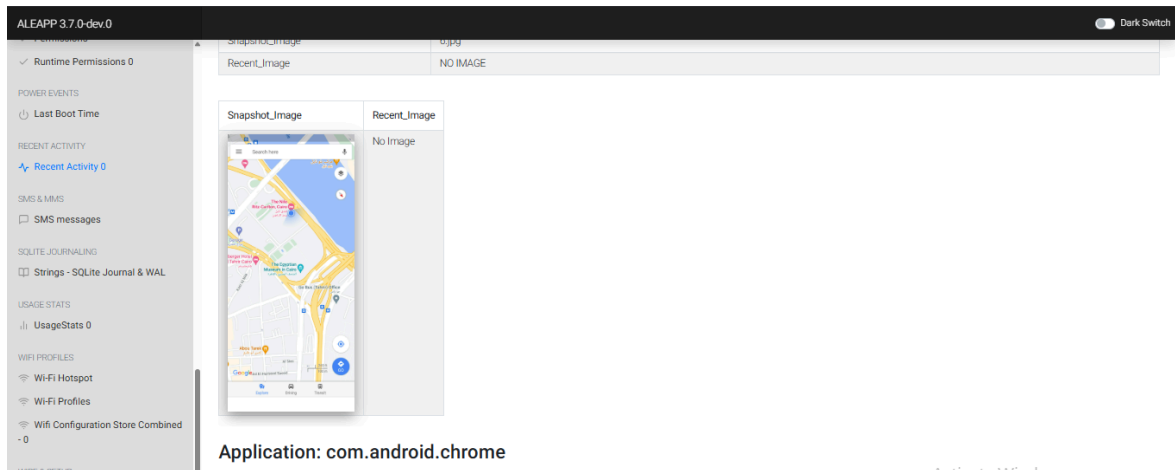
flag: Shady Wahab

Q4

Based on the statement from the victim's family, they said that on **September 20, 2023** , he departed from his residence without informing anyone of his destination. Where was the victim located at that moment?

## process :

Based on recent activity, the victim had taken a snapshot of their location on Google Maps.



## flag: The Nile Ritz-Carlton

### Q5

The detective continued his investigation by questioning the hotel lobby. She informed him that the victim had reserved the room for 10 days and had a flight scheduled thereafter. The investigator believes that the victim may have stored his ticket information on his phone. Look for where the victim intended to travel.

## process:

We found an image of the victim's ticket saved in Google Photos, which appears to be a flight ticket to Las Vegas.

ALEAPP 3.7.0 dev.0

Dark Switch

FIREBASE CLOUD MESSAGING

FCM-Dump.com discord

GOOGLE PHOTOS

Google Photos (gphotos-1) - Cache

IMAGE MANAGER CACHE

Image Manager Cache

INSTALLED APPS

App Icon

Google Play Links for Apps

Installed Apps (GMS) for user 0

Packages

PERMISSIONS

Appops.xml

Package and Shared User

Permission Trees

Permissions

Runtime Permissions 0

ALEAPP

Google Photos (gphotos-1) - Cache report

Total number of entries: 5

Google Photos (gphotos-1) - Cache located at: C:\Users\loki\Desktop\138-The-Crime\ALEAPP\_Reports\_2026-06-09\_Tuesday\_112938\data\data\data\com.google.android.apps.photos\database\disk\_cache

Show 15 entries

Search:

| Timestamp           | Key                                                               | Image | Size   | Pending Deletion |
|---------------------|-------------------------------------------------------------------|-------|--------|------------------|
| 2023-09-19 23:06:07 | 55187f283e17f99c1eacae950020761c04fa32579be38f10880ea7b3ebc7d6    |       | 158508 |                  |
| 2023-09-19 23:06:07 | 6a0762d4cd7283ac28a2f0b635436c9737815e2a3a2b8abcf591813df9b1329   |       | 149811 |                  |
| 2023-09-19          | 13775a7d40bcc820f4cf14a663d38c5f8f9669bac0a21761d55b3851fb0413528 |       | 20155  |                  |

EGYPT AIRLINES

592 428 778 75 42 33

Passenger

MOHAMED AHMED

Boarding Time

09.00 AM

Gate

08

Flight

310

Date

01.10.2023

From

Cairo

To

Las Vegas

Seat

20

Group

C

C CLASS

Please watch the departure board for the boarding & gate update boarding ends 15 min before departure

Passenger

MOHAMED AHMED

Boarding Time

09.00 AM

Gate

08

Flight

310

From

Cairo

To

Las Vegas

flag::Las Vegas

Q6

After examining the victim's Discord conversations, we discovered he had arranged to meet a friend at a specific location. Can you determine where this meeting was supposed to occur?

process :

Based on the conversation on Discord, we identified the arranged meeting location discussed by the individuals.

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ALEAPP 3.7.0-dev0

Partner Settings

SIM info 0

Settings Secure 0

DEVICE INFORMATION

ADB Hosts

OS Version

DISCORD CHATS

Discord Chats

EMULATED STORAGE METADATA

Emulated Storage Metadata - Files

FIREBASE CLOUD MESSAGING

FCM-Dump-com.discord

GOOGLE PHOTOS

Google Photos (photos-1) - Cache

IMAGE MANAGER CACHE

Image Manager Cache

INSTALLED APPS

Dark Switch

| Timestamp                        | Channel ID          | ID                  | Username  | Content                                                                                                                                   | Filename | URL | Proxy URL | Mentions | Roles | Pinned |
|----------------------------------|---------------------|---------------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------|----------|-----|-----------|----------|-------|--------|
| 2023-09-20T00:57:26.406000+00:00 | 1153848030269804606 | 1153857419345137687 | inferno_o | Hey mate<br>Some changes have occurred in the plan. I have booked my travel ticket for 01/10 at 9:00 AM. Where am I supposed to meet you? |          |     |           |          |       | False  |
| 2023-09-20T20:46:02.237000+00:00 | 1153848030269804606 | 1154156539620376576 | rob1rs0n. | What a wonderful news! We'll meet at **The Mob Museum**, I'll await your call when you arrive. Enjoy you flight bro                       |          |     |           |          |       | False  |

Activate Windows

Go to Settings to activate Windows.

flag: The Mob Museum